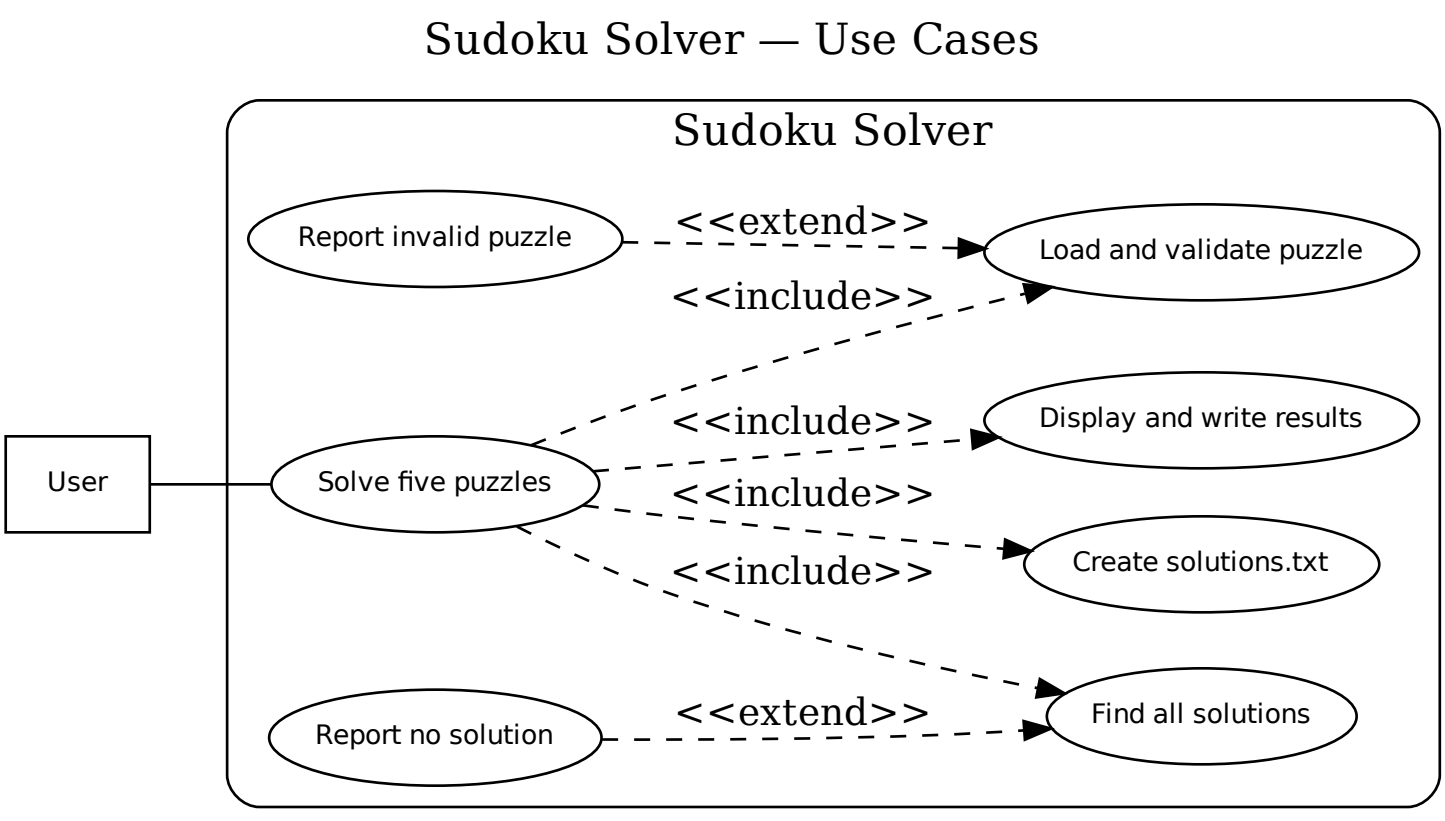
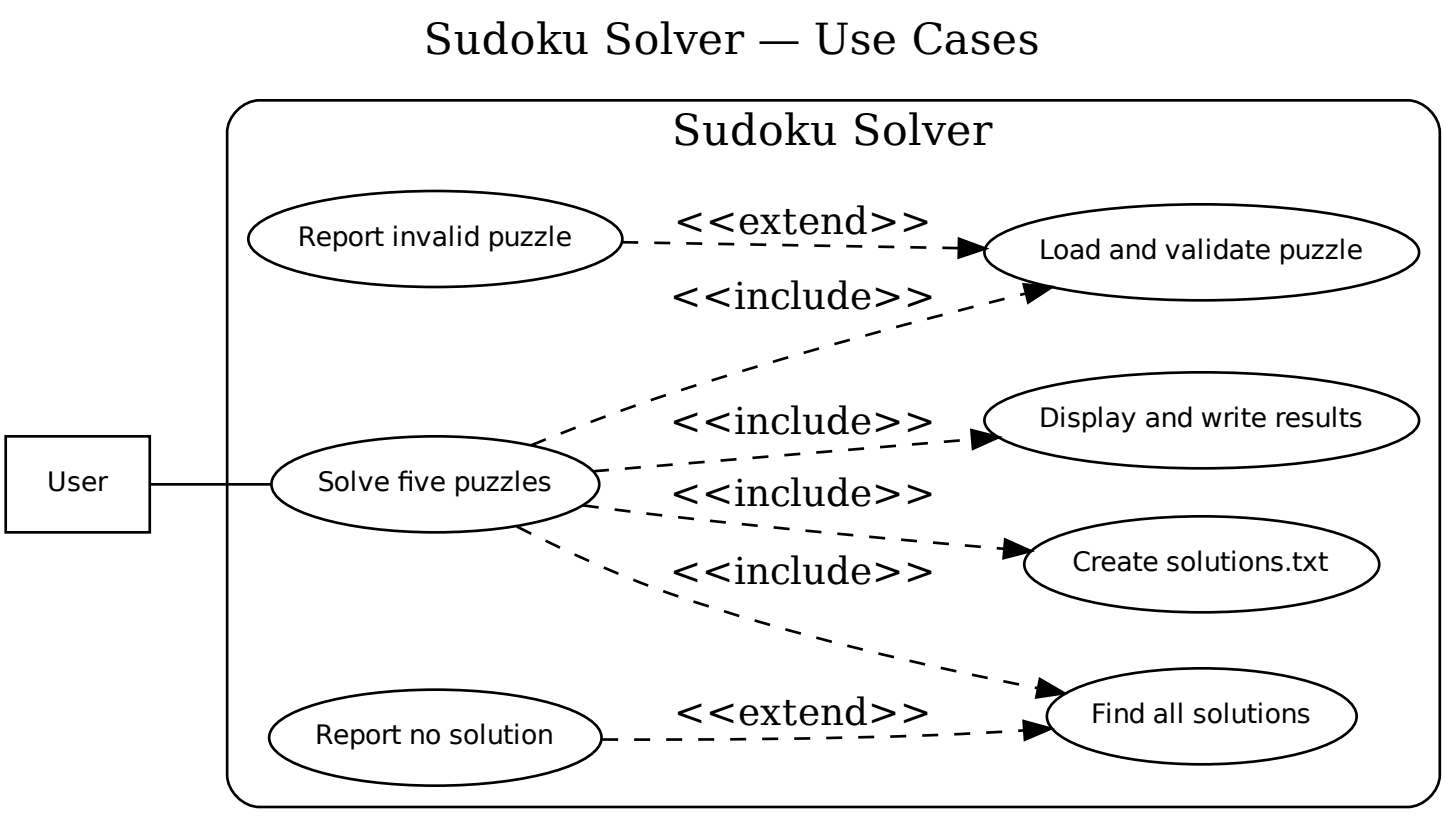

Sudoku Solver — Use Cases
Sudoku Solver
Report invalid puzzle
<<extend>>
Load and validate puzzle
<<include>>
<<include>>
Display and write results
User
Solve five puzzles
<<include>>
Create solutions.txt
<<include>>
Find all solutions
Report no solution
<<extend>>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Assignment 6 — UML Use Case

Sudoku Solver: combined solutions output

Primary actor: User. Main use case: Solve five Sudoku puzzles.

Brief description

The user runs Assignment6.exe in a directory with five puzzle files. The system loads and validates each puzzle, enumerates every solution using recursive backtracking, prints results, and writes all results into a single solutions.txt file.

Event flow

1. User launches the program.
2. System creates solutions.txt.
3. For N from 1 through 5, system writes Puzzle: puzzleN.txt.
4. System loads and validates the 9×9 input.
5. If valid, system recursively finds all solutions and displays/writes each grid.
6. System writes Solutions found: n; if none or invalid, writes No solution found.
7. System proceeds to the next puzzle and closes the file at exit.

Detailed use case

Preconditions: executable and input files are available in the current directory, and output directory is writable. Inputs contain exactly 81 whitespace-separated tokens (digits 1–9 or underscore). Success postcondition: solutions.txt has five labeled sections with all valid solutions and counts. Alternate A1: missing, malformed, or contradictory puzzle produces No solution found. and processing continues. Alternate A2: a valid but unsolvable puzzle produces No solution found. Alternate A3: multiple solutions are all enumerated and written. Alternate A4: failure to open solutions.txt causes an error message and nonzero exit.

Relationships

Solve five puzzles includes: Create combined output, Load/validate puzzle, Search recursively, Display/write results. Report invalid input extends Load/validate puzzle. Report no solution extends Search recursively.